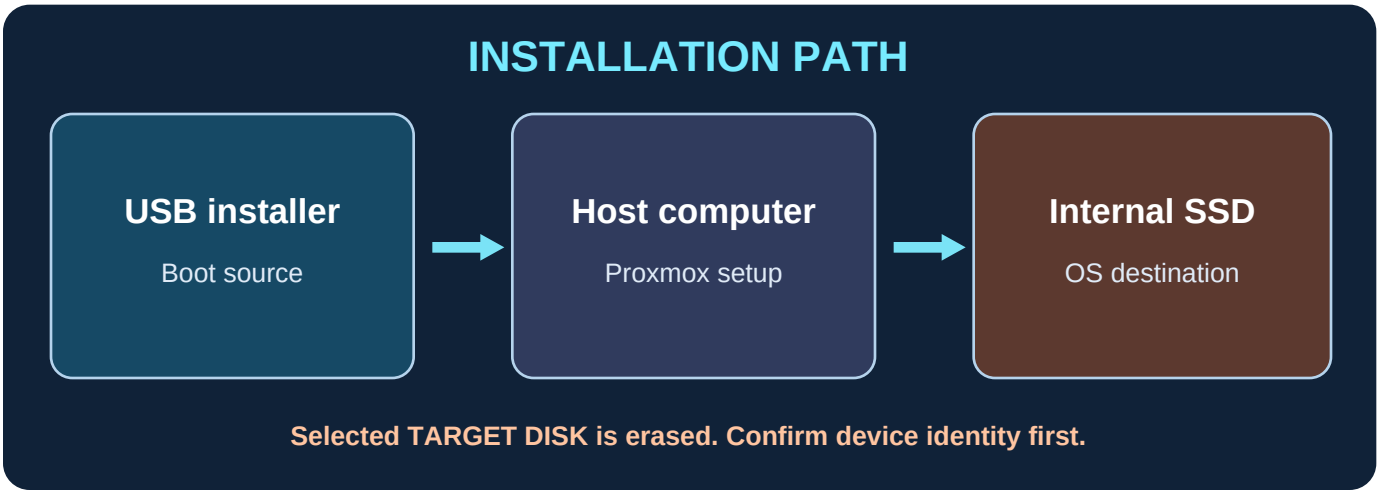
LAB 02: Hardware Preparation & Installation

Network Support Services | Guided setup study guide | Created by Mr. Beaird

Mission: Prepare a physical host, boot the installer correctly, choose the internal OS drive, configure secure administration and a static management network, then verify the Proxmox web interface. This guide supplements the interactive Lab 02 HTML.



Learning outcomes

• Identify installation USB versus internal OS SSD and explain the consequences of selecting the wrong disk.

• Locate hardware virtualization settings (Intel VT-x / AMD-V / SVM) in BIOS or UEFI.

• Plan a valid FQDN, static management IPv4 address, prefix, gateway and DNS server.

• Explain each installer page and validate browser access using HTTPS on port 8006.

Safety gate: Installation wipes the chosen target disk. The HTML simulator does not access or erase real equipment. Obtain instructor approval before physical installation.

1 | Hardware and firmware preflight

Item	Technician checkpoint
CPU	64-bit compatible processor; Intel VT-x or AMD-V enabled for KVM.
Memory	Adequate host RAM plus additional capacity for planned guests.
OS disk	Confirmed internal SSD, ideally 128 GB for this classroom plan.
Networking	Active wired NIC and known switch port / network.
Boot source	Prepared USB installer with Proxmox VE ISO; USB is not OS destination.
Firmware	Appropriate boot mode and correct one-time boot device.

For a testing installation, Proxmox documents lower minimums than for production. Plan RAM for guest workloads; do not assume the minimal evaluation configuration is sufficient for classroom projects.

2 | Prepare trusted installation media

• Download the Proxmox VE ISO only from the official Proxmox download page.

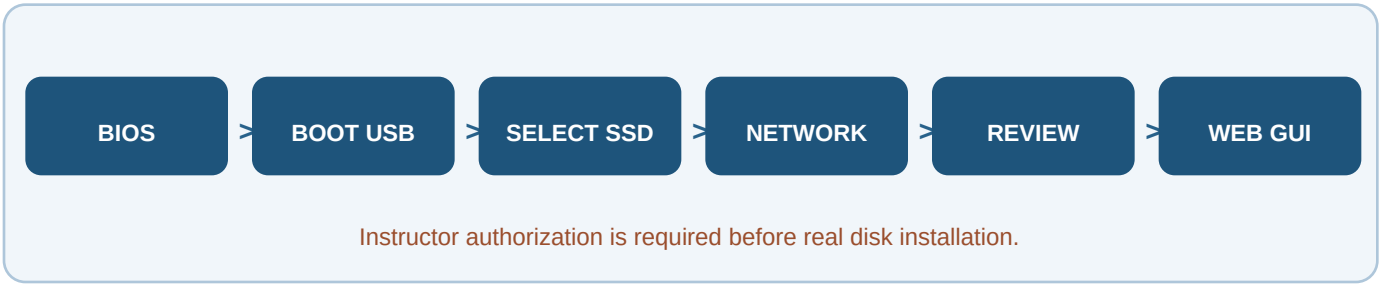
• Compare the downloaded ISO SHA-256 checksum against the published value.

• Use an approved USB writing tool to write the ISO image to a dedicated USB flash drive.

• Label the USB clearly and identify the internal SSD before booting.

Official resources: www.proxmox.com/en/downloads and pve.proxmox.com/pve-docs/

3 | Installation sequence



The GUI installer generally asks for the license agreement, target disk, country/time zone/keyboard, administrator credentials and email, then management networking before reviewing the final summary. Labels can vary slightly between releases.

4 | Target-disk selection: prevent data loss

Training device	Purpose	Correct action
/dev/sdb - 32 GB USB	Boot / installation media	Never choose as the OS target in this scenario
/dev/sda - 128 GB SSD	Internal operating-system drive	Select only after confirming real disk identity
/dev/sdc - 2 TB disk	Existing data drive	Preserve; do not overwrite

Linux device paths are illustrative. Device names can change between machines and boots. Check size, model and serial rather than assuming the internal drive always appears as /dev/sda.

5 | Management network worksheet

Node	Hostname / FQDN	IPv4 / Prefix
PVE1	pve1.lab.local	192.168.254.10/24
PVE2	pve2.lab.local	192.168.254.20/24
PVE3	pve3.lab.local	192.168.254.30/24

Default classroom example: gateway 192.168.254.1; DNS 192.168.254.1. Verify that these are correct for your actual isolated subnet. A /24 corresponds to 255.255.255.0.

FQDN: a fully qualified domain name combines host and domain, such as pve1.lab.local. **Gateway:** a router address used to reach other networks. **DNS:** resolves names to addresses.

6 | Security and post-reboot access

• Set a unique strong root password in the real installer; do not include it in a PDF report.

• Use a valid administrator email and the correct local time zone.

• After the install completes, remove USB media if prompted and boot from the SSD.

• Open https://<management-IP>:8006 in a supported browser.

• A browser certificate warning can occur before installing a trusted internal CA. Record it; internal CA configuration is covered in a later lab.

7 | Troubleshooting decision tree

Symptom	Investigation
Installer does not boot	Check ISO USB writing, UEFI boot options, boot selection and media integrity.
No target disk displayed	Verify firmware storage settings, drive connection, model and hardware compatibility.
VM guests cannot use KVM	Check BIOS VT-x / AMD-V and avoid unsupported nested virtualization settings.
Browser cannot reach :8006	Check host boot, physical link, correct NIC, IPv4/prefix, gateway, DNS and same-subnet reachability.
Certificate not trusted	Expected on some default installations; use a trusted internal CA in a later lab.

8 | Student completion checklist

■ Identified USB and internal SSD separately

■ Verified hardware virtualization setting

■ Recorded FQDN and management network values

■ Completed all seven HTML installer simulation checkpoints

■ Scored at least 80% on the knowledge assessment

■ Recorded technician evidence and submitted printed PDF report

Vocabulary

Hypervisor: platform that hosts virtual machines. **ISO:** disk image used as installation media. **UEFI:** firmware interface used at startup. **VT-x / AMD-V:** CPU virtualization extensions. **FQDN:** full host and DNS domain name. **CIDR:** prefix notation such as /24. **Gateway:** router used to reach other networks. **Root:** privileged administrator account. **HTTPS:** encrypted browser protocol. **Port 8006:** default Proxmox web management port.

Documentation and references

Proxmox VE System Requirements: https://www.proxmox.com/en/products/proxmox-virtual-environment/requirements

Proxmox Installation Documentation: https://github.com/proxmox/pve-docs/blob/master/pve-installation.adoc

Proxmox VE Administration Guide: https://pve.proxmox.com/pve-docs/pve-admin-guide.pdf